

23122931 S3 Bucket Step-by-Step Guide

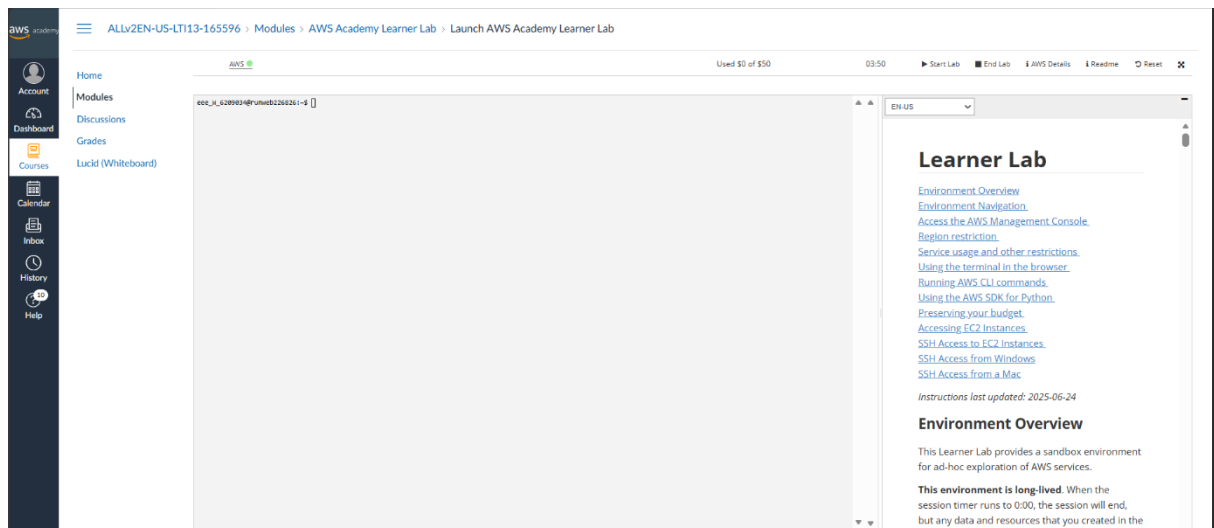
Introduction

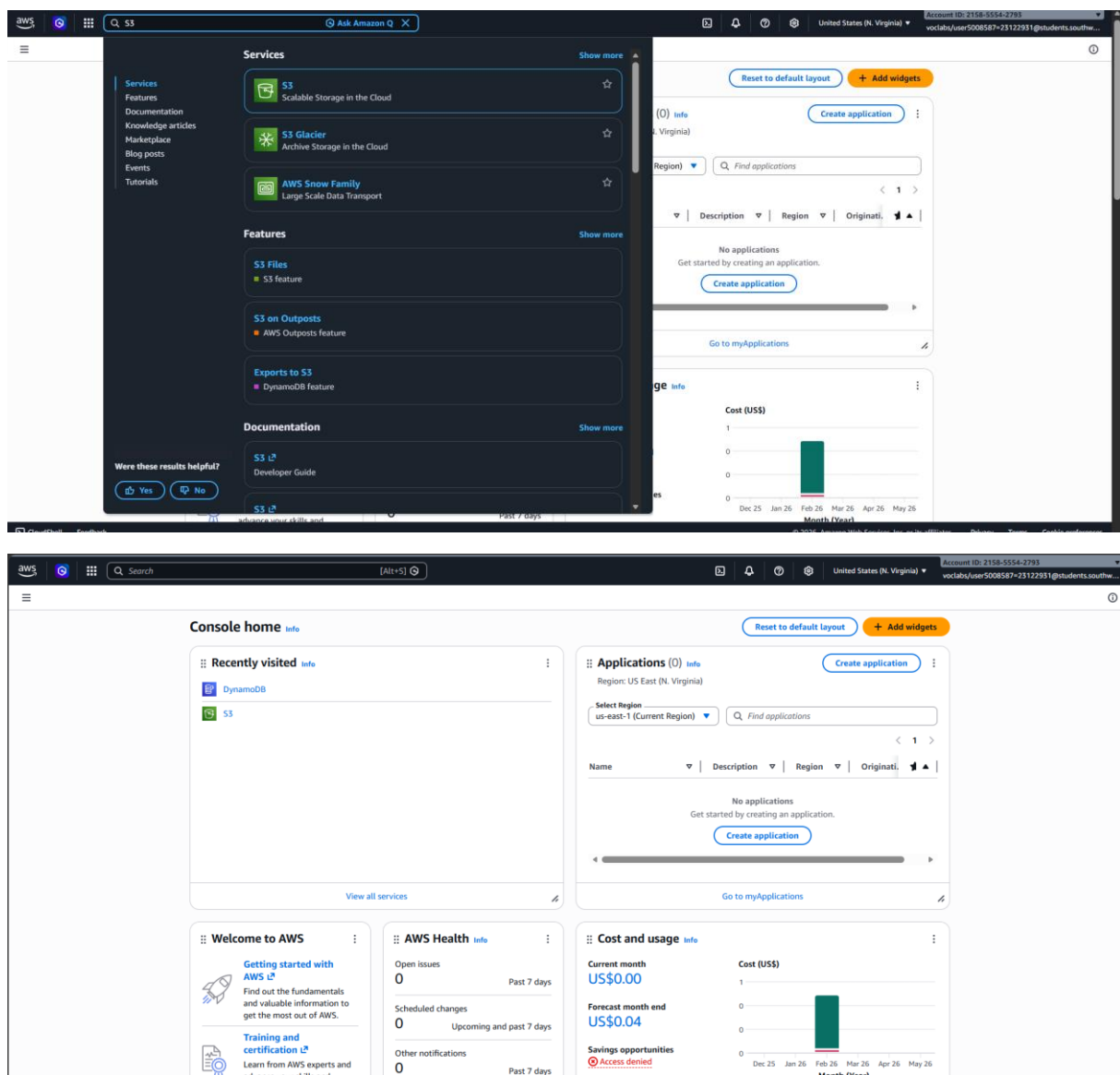
This document guides you through creating and configuring the Amazon S3 bucket needed for the EcoStream Energy cloud storage solution. The guide describes how to create the bucket “23122931-ecostream-energy-bucket” in Amazon Web Services (AWS), including the settings applied to make secure and reliable cloud-based data storage possible.

The page also displays the successful upload of the given telemetry collection into the S3 bucket for future database integration and querying. The tutorial includes relevant screenshots and configuration explanations throughout to help EcoStream staff with limited expertise of cloud computing to successfully repeat the setup process. I signed into the AWS Academy Learner Lab environment and began the work of configuring cloud storage for EcoStream Energy. The learner lab environment gave temporary access to the AWS services and resources I will need to build and operate the S3 storage architecture. To start the configuration procedure, the following steps were used:

Step 1: After successfully entering the AWS Management Console from the learner lab environment, the AWS search bar was utilized to search for "S3". Then I selected the Amazon S3 service under category “Services” to start the configuration of the cloud storage bucket needed to store the EcoStream telemetry data provided in the assessment resource.

This step was an important step as Amazon S3 provides scalable object storage, which would later be utilized to store the “energy-telemetry-2025-2026-dbjson” dataset before integrating it with DynamoDB for querying and analysis.





Step 2: To set up the cloud storage environment required for the EcoStream dataset, the Amazon S3 bucket creation page was opened from the S3 service dashboard. In the “General configuration” section, the AWS Region “US East (N. Virginia) us-east-1” was kept so that it would be the same as that utilized across the assessment brief.

The “General purpose” bucket type was chosen because it is suitable for common cloud storage operations and offers high availability across many Availability Zones. The bucket name “23122931-ecostream-energy-bucket ” is given in the assessment brief.

The default “ACLs disabled” object ownership setting was retained to simplify access management using bucket policies instead of legacy access control lists. Furthermore, the “Block all public access” setting was still enabled to increase the security of the bucket and prevent unauthorised public access to the stored telemetry dataset.

The “bucket versioning” was disabled because the assessment did not require the ability to save multiple object versions, only required the fundamental dataset storage feature. For the encryption settings, the default server-side encryption option “SSE-S3” was kept to automatically encrypt all uploaded files stored in the bucket using Amazon S3-managed encryption keys.

These parameters were chosen to establish a secure, well designed and cost effective cloud storage environment for the EcoStream telemetry dataset in line with AWS best security practices.

The screenshot shows the 'Create bucket' page in the AWS console, specifically the 'General configuration' section. The page is for the 'us-east-1' region. The 'Bucket type' is set to 'General purpose'. The 'Bucket namespace' is set to 'Global namespace'. The 'Bucket name' is '23122931-ecostream-energy-bucket'. The 'Copy settings from existing bucket' option is not selected. The 'Choose bucket' button is visible at the bottom.

Create bucket [info](#)

Buckets are containers for data stored in S3.

General configuration

AWS Region
US East (N. Virginia) us-east-1

Bucket type [info](#)

☒ **General purpose**
Recommended for most use cases and access patterns. General purpose buckets are the original S3 bucket type. They allow a mix of storage classes that redundantly store objects across multiple Availability Zones.

☐ **Directory**
Recommended for low-latency use cases. These buckets use only the S3 Express One Zone storage class, which provides faster processing of data within a single Availability Zone.

Bucket namespace
Choose the namespace where you want to create your bucket. [Learn more](#)

☒ **Global namespace**
By default, S3 creates general purpose buckets in the global namespace.

☐ **Account Regional namespace (recommended)**
General purpose buckets created in your account Regional namespace are unique to your account. These buckets can never be created by another AWS account.

Bucket name [info](#)

23122931-ecostream-energy-bucket

Bucket names must be 3 to 63 characters and unique within the global namespace. Bucket names must also begin and end with a letter or number. Valid characters are a-z, 0-9, periods (.), and hyphens (-). [Learn more](#)

Copy settings from existing bucket - optional
Only the bucket settings in the following configuration are copied.

Enmar - s3 - /bucket/energy

CloudShell Feedback © 2026, Amazon Web Services, Inc. or its affiliates. Privacy Terms Cookie preferences

The screenshot shows the 'Create bucket' page in the AWS console, specifically the 'Object Ownership' and 'Block Public Access settings' sections. The 'Object Ownership' section has 'ACLs disabled (recommended)' selected. The 'Block Public Access settings' section has 'Block all public access' checked, and all four sub-settings are also checked.

Object Ownership [info](#)

Control ownership of objects written to this bucket from other AWS accounts and the use of access control lists (ACLs). Object ownership determines who can specify access to objects.

Object Ownership

☒ **ACLs disabled (recommended)**
All objects in this bucket are owned by this account. Access to this bucket and its objects is specified using only policies.

☐ **ACLs enabled**
Objects in this bucket can be owned by other AWS accounts. Access to this bucket and its objects can be specified using ACLs.

Object Ownership
Bucket owner enforced

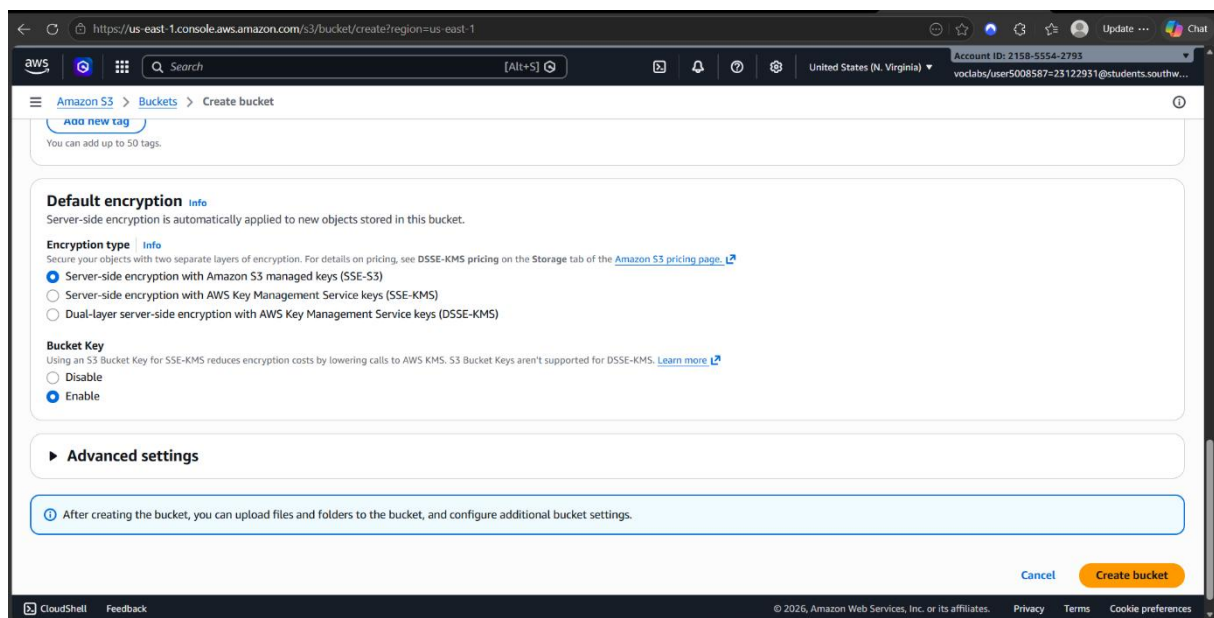
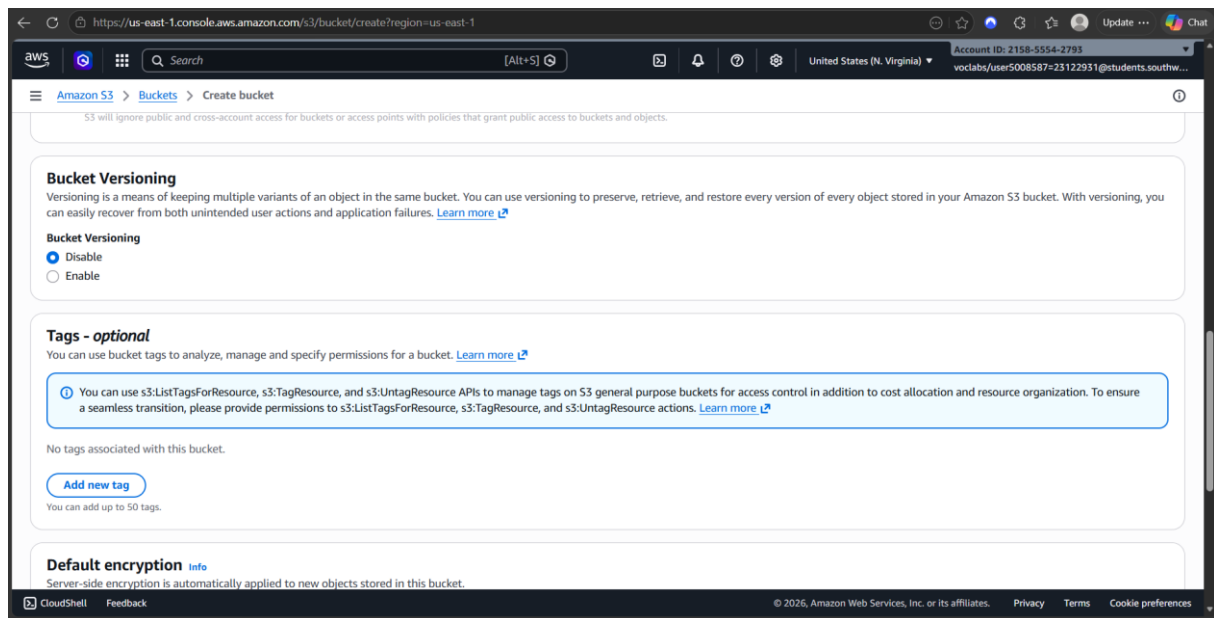
Block Public Access settings for this bucket

Public access is granted to buckets and objects through access control lists (ACLs), bucket policies, access point policies, or all. In order to ensure that public access to this bucket and its objects is blocked, turn on Block all public access. These settings apply only to this bucket and its access points. AWS recommends that you turn on Block all public access, but before applying any of these settings, ensure that your applications will work correctly without public access. If you require some level of public access to this bucket or objects within, you can customize the individual settings below to suit your specific storage use cases. [Learn more](#)

☒ **Block all public access**
Turning this setting on is the same as turning on all four settings below. Each of the following settings are independent of one another.

- ☒ **Block public access to buckets and objects granted through new access control lists (ACLs)**
S3 will block public access permissions applied to newly added buckets or objects, and prevent the creation of new public access ACLs for existing buckets and objects. This setting doesn't change any existing permissions that allow public access to S3 resources using ACLs.
- ☒ **Block public access to buckets and objects granted through any access control lists (ACLs)**
S3 will ignore all ACLs that grant public access to buckets and objects.
- ☒ **Block public access to buckets and objects granted through new public bucket or access point policies**
S3 will block new bucket and access point policies that grant public access to buckets and objects. This setting doesn't change any existing policies that allow public access to S3 resources.
- ☒ **Block public and cross-account access to buckets and objects through any public bucket or access point policies**
S3 will ignore public and cross-account access for buckets or access points with policies that grant public access to buckets and objects.

CloudShell Feedback © 2026, Amazon Web Services, Inc. or its affiliates. Privacy Terms Cookie preferences



Step 3: After setting up the bucket configuration, the option “Create bucket” was selected to deploy the storage bucket. AWS successfully created the bucket “23122931-ecostream-energy-bucket”. And the bucket dashboard opened automatically.

The bucket interface had the Objects, Properties, Permissions, Metrics, and Management tabs proving the successful provisioning of the storage environment ready to store data. This phase verified that the S3 bucket infrastructure required for the evaluation has been successfully set up.

The “Upload” option inside the S3 bucket dashboard was selected to upload the EcoStream telemetry dataset into the newly created storage bucket. The dataset file “energy-telemetry-2025-2026-dbjson” was uploaded successfully into the bucket storage environment.

The upload status screen indicated that the upload was completed successfully and no mistakes were detected. The uploaded file was listed in the files and folders area with its file type and storage size, which signified that the dataset was successfully stored in the Amazon S3 bucket. This step was necessary because according to the assessment brief, the telemetry dataset has to be saved in Amazon S3 first, and then imported into DynamoDB for database analysis and querying operations.

This step was necessary because the assessment required the telemetry dataset to be stored inside Amazon S3 before it could later be imported into DynamoDB for querying and database analysis tasks.

